

## Apply filters to SQL queries

### Project description

Databases contain large amounts of data organized by tables. To access database data efficiently, search query language (SQL) is used to quickly search data and return information. Specific syntax is necessary to target data that is time consuming to find. Using SQL filter syntax such as WHERE, LIKE, OR, AND, and NOT helps searches to minimize the amount of data in the table and find what information is needed. In this project, I search through a database for specific information in a sample organization for employee and login information.

### Retrieve after hours failed login attempts

My team discovered a security incident that occurred after business hours which can potentially be a brute force attempt. In this screenshot below, I use SQL filters to query the organization's database for failed login attempts. To search through all data I use **SELECT \*** and FROM log\_in\_attempts to source the data from the specific information logs in the database. I use the **WHERE** argument to search for the after hour login time which is designated to be **18:00** or 6 o'clock. I use the **AND** to add an additional argument and use **= 0** to signal that logins were unsuccessful.

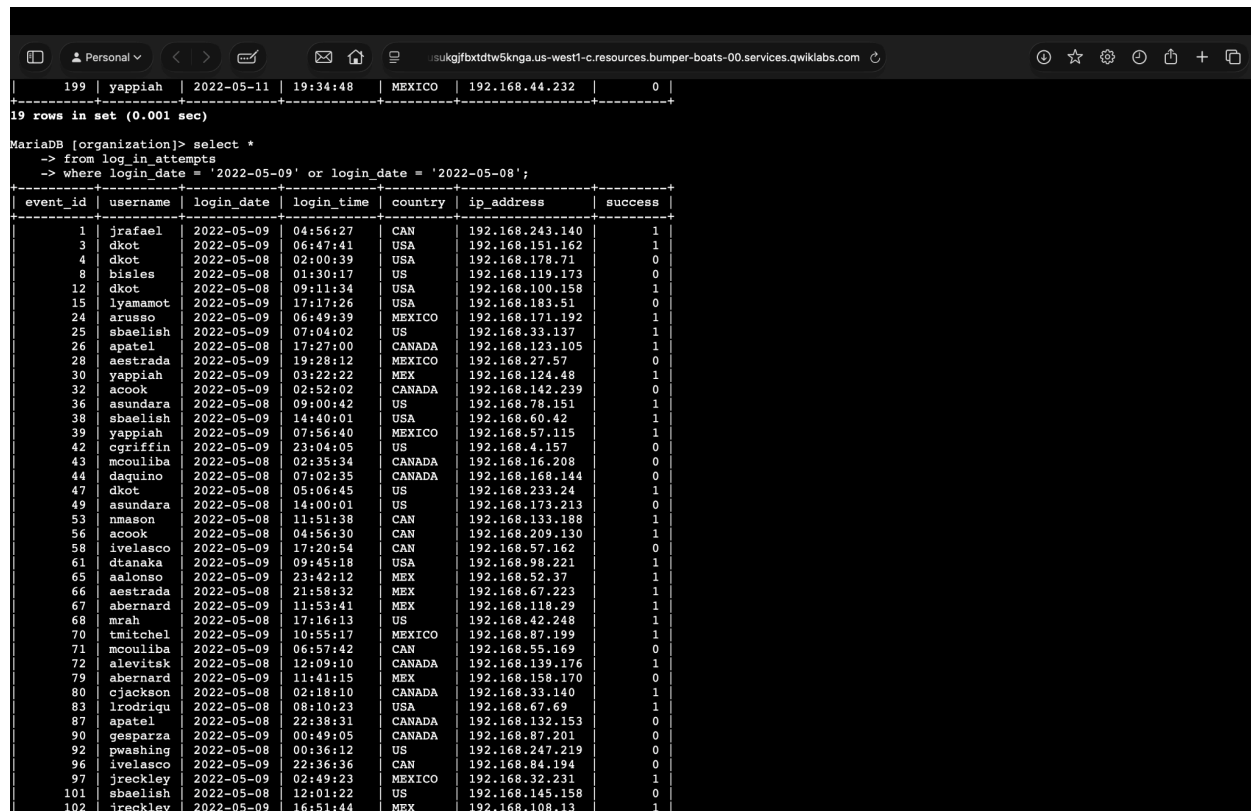
```
105 | c.jackson | 2022-05-12 | 19:36:42 | CAN | 192.168.247.153 | 1
107 | bisles | 2022-05-12 | 20:25:57 | USA | 192.168.116.187 | 0
108 | daquino | 2022-05-09 | 21:30:48 | CANADA | 192.168.15.110 | 1
111 | aestrada | 2022-05-10 | 22:00:26 | MEXICO | 192.168.76.27 | 0
115 | ivelasco | 2022-05-10 | 23:06:01 | CAN | 192.168.154.1 | 1
116 | tmitchel | 2022-05-10 | 20:33:27 | MEXICO | 192.168.119.26 | 1
118 | smartell | 2022-05-12 | 23:21:31 | MEXICO | 192.168.173.196 | 1
119 | tmitchel | 2022-05-11 | 23:07:13 | MEXICO | 192.168.110.175 | 1
121 | btang | 2022-05-10 | 22:00:36 | US | 192.168.80.143 | 1
126 | jrafael | 2022-05-12 | 18:47:52 | CAN | 192.168.22.16 | 1
127 | abellmas | 2022-05-09 | 21:20:51 | CANADA | 192.168.70.122 | 0
131 | bisles | 2022-05-09 | 20:03:55 | US | 192.168.113.171 | 0
132 | rjensen | 2022-05-12 | 23:26:03 | MEX | 192.168.9.166 | 1
155 | cgriffin | 2022-05-12 | 22:18:42 | USA | 192.168.236.176 | 0
158 | smartell | 2022-05-09 | 19:30:32 | MEXICO | 192.168.190.178 | 1
160 | jclark | 2022-05-10 | 20:49:00 | CANADA | 192.168.214.49 | 0
164 | jclark | 2022-05-12 | 21:15:52 | CAN | 192.168.18.34 | 1
173 | asundara | 2022-05-12 | 23:17:52 | US | 192.168.58.217 | 1
199 | yappiah | 2022-05-11 | 19:34:48 | MEXICO | 192.168.44.232 | 0
39 rows in set (0.001 sec)

MariaDB [organization]> select * from log_in_attempts where login_time > '18:00' and success = 0;
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 | 0
| 18 | pwashing | 2022-05-11 | 19:28:50 | US | 192.168.66.142 | 0
| 20 | tshah | 2022-05-12 | 18:56:36 | MEXICO | 192.168.109.50 | 0
| 28 | aestrada | 2022-05-09 | 19:28:12 | MEXICO | 192.168.27.57 | 0
| 34 | drosas | 2022-05-11 | 21:02:04 | US | 192.168.45.93 | 0
| 42 | cgriffin | 2022-05-09 | 23:04:05 | US | 192.168.4.157 | 0
| 52 | c.jackson | 2022-05-10 | 22:07:07 | CAN | 192.168.58.27 | 0
| 69 | wjaffrey | 2022-05-11 | 19:55:15 | USA | 192.168.100.17 | 0
| 82 | abernard | 2022-05-12 | 23:38:46 | MEX | 192.168.234.49 | 0
| 87 | apatel | 2022-05-08 | 22:38:31 | CANADA | 192.168.132.153 | 0
| 96 | ivelasco | 2022-05-09 | 22:36:36 | CAN | 192.168.84.194 | 0
| 104 | asundara | 2022-05-11 | 18:38:07 | US | 192.168.96.200 | 0
| 107 | bisles | 2022-05-12 | 20:25:57 | USA | 192.168.116.187 | 0
| 111 | aestrada | 2022-05-10 | 22:00:26 | MEXICO | 192.168.76.27 | 0
| 127 | abellmas | 2022-05-09 | 21:20:51 | CANADA | 192.168.70.122 | 0
| 131 | bisles | 2022-05-09 | 20:03:55 | US | 192.168.113.171 | 0
| 155 | cgriffin | 2022-05-12 | 22:18:42 | USA | 192.168.236.176 | 0
| 160 | jclark | 2022-05-10 | 20:49:00 | CANADA | 192.168.214.49 | 0
| 199 | yappiah | 2022-05-11 | 19:34:48 | MEXICO | 192.168.44.232 | 0
19 rows in set (0.001 sec)

MariaDB [organization]>
```

## Retrieve login attempts on specific dates

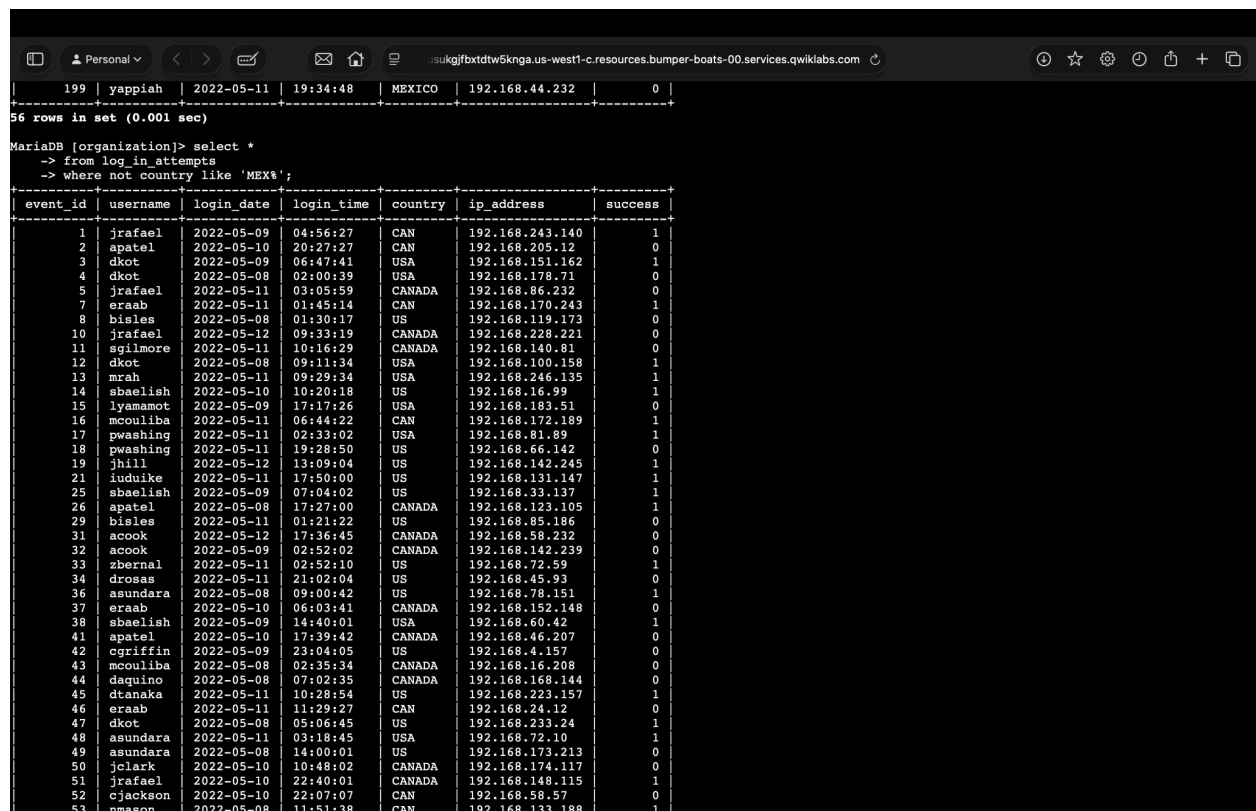
A suspicious event occurred on the date 2022-05-09. To investigate the event, login attempts for the day of the event and the day before should be reviewed. In this screenshot below, I queried for login attempts on specific dates. Again I used **SELECT \*** to access data categories and **FROM log\_in\_attempts** for login information. Then, I searched **WHERE login\_date = '2022-05-09'** and the **OR** argument for **login\_date '2022-05-08'**. This gave all data for the both dates inside of the SQL search while showing whether they were successful or not.



| 199                                                                                                                             | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  | 0       |
|---------------------------------------------------------------------------------------------------------------------------------|----------|------------|------------|---------|-----------------|---------|
| 19 rows in set (0.001 sec)                                                                                                      |          |            |            |         |                 |         |
| MariaDB [organization]> select *<br>-> from log_in_attempts<br>-> where login_date = '2022-05-09' or login_date = '2022-05-08'; |          |            |            |         |                 |         |
| event_id                                                                                                                        | username | login_date | login_time | country | ip_address      | success |
| 1                                                                                                                               | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 | 1       |
| 3                                                                                                                               | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 | 1       |
| 4                                                                                                                               | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  | 0       |
| 8                                                                                                                               | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 | 0       |
| 12                                                                                                                              | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 | 1       |
| 15                                                                                                                              | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  | 0       |
| 24                                                                                                                              | arusso   | 2022-05-09 | 06:49:39   | MEXICO  | 192.168.171.192 | 1       |
| 25                                                                                                                              | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  | 1       |
| 26                                                                                                                              | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 | 1       |
| 28                                                                                                                              | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   | 0       |
| 30                                                                                                                              | yappiah  | 2022-05-09 | 03:22:22   | MEX     | 192.168.124.48  | 1       |
| 32                                                                                                                              | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 | 0       |
| 36                                                                                                                              | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  | 1       |
| 38                                                                                                                              | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   | 1       |
| 39                                                                                                                              | yappiah  | 2022-05-09 | 07:56:40   | MEXICO  | 192.168.57.115  | 1       |
| 42                                                                                                                              | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 43                                                                                                                              | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  | 0       |
| 44                                                                                                                              | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 | 0       |
| 47                                                                                                                              | dkot     | 2022-05-08 | 05:06:45   | US      | 192.168.233.24  | 1       |
| 49                                                                                                                              | asundara | 2022-05-08 | 14:00:01   | US      | 192.168.173.213 | 0       |
| 53                                                                                                                              | nmason   | 2022-05-08 | 11:51:38   | CAN     | 192.168.133.188 | 1       |
| 56                                                                                                                              | acook    | 2022-05-08 | 04:56:30   | CAN     | 192.168.209.130 | 1       |
| 58                                                                                                                              | ivelasco | 2022-05-09 | 17:20:54   | CAN     | 192.168.57.162  | 0       |
| 61                                                                                                                              | dtanaka  | 2022-05-09 | 09:45:18   | USA     | 192.168.98.221  | 1       |
| 65                                                                                                                              | aalonso  | 2022-05-09 | 23:42:12   | MEX     | 192.168.52.37   | 1       |
| 66                                                                                                                              | aestrada | 2022-05-08 | 21:58:32   | MEX     | 192.168.67.223  | 1       |
| 67                                                                                                                              | abernard | 2022-05-09 | 11:53:41   | MEX     | 192.168.118.29  | 1       |
| 68                                                                                                                              | mrab     | 2022-05-08 | 17:16:13   | US      | 192.168.42.248  | 1       |
| 70                                                                                                                              | tmitchel | 2022-05-09 | 10:55:17   | MEXICO  | 192.168.87.199  | 1       |
| 71                                                                                                                              | mcouliba | 2022-05-09 | 06:57:42   | CAN     | 192.168.55.169  | 0       |
| 72                                                                                                                              | alevitsk | 2022-05-08 | 12:09:10   | CANADA  | 192.168.139.176 | 1       |
| 79                                                                                                                              | abernard | 2022-05-09 | 11:41:15   | MEX     | 192.168.158.170 | 0       |
| 80                                                                                                                              | cjackson | 2022-05-08 | 02:18:10   | CANADA  | 192.168.33.140  | 1       |
| 83                                                                                                                              | lrodriqu | 2022-05-08 | 08:10:23   | USA     | 192.168.67.69   | 1       |
| 87                                                                                                                              | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 | 0       |
| 90                                                                                                                              | gesparza | 2022-05-09 | 00:49:05   | CANADA  | 192.168.87.201  | 0       |
| 92                                                                                                                              | psashing | 2022-05-08 | 00:36:12   | US      | 192.168.247.219 | 0       |
| 96                                                                                                                              | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  | 0       |
| 97                                                                                                                              | jreckley | 2022-05-09 | 02:49:23   | MEXICO  | 192.168.32.231  | 1       |
| 101                                                                                                                             | sbaelish | 2022-05-08 | 12:01:22   | US      | 192.168.145.158 | 0       |
| 102                                                                                                                             | jreckley | 2022-05-09 | 16:51:44   | MEX     | 192.168.108.13  | 1       |

## Retrieve login attempts outside of Mexico

In this search, I searched for suspicious activity occurring outside of Mexico. To lookup this data I used **SELECT \*** and **FROM log\_in\_attempts**. Then, I used the WHERE argument to search for countries that aren't Mexico. I added the argument **NOT country LIKE 'MEX%'**; The **NOT** argument negates the countries search which returns the opposite of the argument. The **LIKE** argument allows for searches that may contain the string starting with **'MEX'**. The **%** is positioned after the **MEX** in case that the country only uses the first 3 letters or the entire name depending on the spelling.



```
199 | yappiah | 2022-05-11 | 19:34:48 | MEXICO | 192.168.44.232 | 0 |
56 rows in set (0.001 sec)

MariaDB [organization]> select *
-> from log_in_attempts
-> where not country like 'MEX%';
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 1        | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 | 1       |
| 2        | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  | 0       |
| 3        | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 | 1       |
| 4        | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  | 0       |
| 5        | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  | 0       |
| 6        | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 | 1       |
| 7        | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 | 0       |
| 10       | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 | 0       |
| 11       | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  | 0       |
| 12       | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 | 1       |
| 13       | mrah     | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 | 1       |
| 14       | sbaelish | 2022-05-10 | 10:20:18   | US      | 192.168.16.99   | 1       |
| 15       | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  | 0       |
| 16       | mcouliba | 2022-05-11 | 06:44:22   | CAN     | 192.168.172.189 | 1       |
| 17       | pwashing | 2022-05-11 | 02:33:02   | USA     | 192.168.81.89   | 1       |
| 18       | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  | 0       |
| 19       | jhill    | 2022-05-12 | 13:09:04   | US      | 192.168.142.245 | 1       |
| 21       | iudulike | 2022-05-11 | 17:50:00   | US      | 192.168.131.147 | 1       |
| 25       | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  | 1       |
| 26       | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 | 1       |
| 29       | bisles   | 2022-05-11 | 01:21:22   | US      | 192.168.85.186  | 0       |
| 31       | acook    | 2022-05-12 | 17:36:45   | CANADA  | 192.168.58.232  | 0       |
| 32       | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 | 0       |
| 33       | zbernal  | 2022-05-11 | 02:52:10   | US      | 192.168.72.59   | 1       |
| 34       | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   | 0       |
| 36       | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  | 1       |
| 37       | eraab    | 2022-05-10 | 06:03:41   | CANADA  | 192.168.152.148 | 0       |
| 38       | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   | 1       |
| 41       | apatel   | 2022-05-10 | 17:39:42   | CANADA  | 192.168.46.207  | 0       |
| 42       | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 43       | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  | 0       |
| 44       | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 | 0       |
| 45       | dtanaka  | 2022-05-11 | 10:28:54   | US      | 192.168.223.157 | 1       |
| 46       | eraab    | 2022-05-11 | 11:29:27   | CAN     | 192.168.24.12   | 0       |
| 47       | dkot     | 2022-05-08 | 05:06:45   | US      | 192.168.233.24  | 1       |
| 48       | asundara | 2022-05-11 | 03:18:45   | USA     | 192.168.72.10   | 1       |
| 49       | asundara | 2022-05-08 | 14:00:01   | US      | 192.168.173.213 | 0       |
| 50       | jelark   | 2022-05-10 | 10:48:02   | CANADA  | 192.168.174.117 | 0       |
| 51       | jrafael  | 2022-05-10 | 22:40:01   | CANADA  | 192.168.148.115 | 1       |
| 52       | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   | 0       |
| 53       | nmason   | 2022-05-08 | 11:51:38   | CAN     | 192.168.133.188 | 1       |

## Retrieve employees in Marketing

My team wanted to perform security updates on specific employee machines in the Marketing department. I was responsible for getting information on these employee machines and will need to query the **employees** table. In this search, I used filters in SQL to create a query that identifies all employees in the Marketing department for all offices in the East building. This SQL filter required **SELECT \*** and **FROM employees** to gather employee information. Then, I added an argument **WHERE department = 'Marketing'**. This argument searches for employees in the **Marketing** department. I added **AND** to join the two arguments together. Finally, the argument **office like 'EAST%'** searches for all offices in the east building. The % after **EAST** ensures that the search begins with east no matter the room number.

```
1063 | 1686m140n569 | lpope | Sales | East-226 |
1069 | NULL | jpark | Finance | East-110 |
1072 | u905v920w694 | esmith | Sales | East-421 |
1074 | w622x645y348 | dcoleman | Information Technology | East-126 |
1075 | x573y883z772 | fbautist | Marketing | East-267 |
1077 | z684u154z359 | ldavis | Human Resources | East-241 |
1082 | z301f659g551 | mjohnson | Information Technology | East-151 |
1085 | h339i498j269 | cperez | Sales | East-325 |
1088 | k865l965m233 | rgosh | Marketing | East-157 |
1095 | r194s893t593 | glopez | Information Technology | East-457 |
1096 | s375t538u194 | kjeffers | Information Technology | East-419 |
1101 | x701y250z303 | ichowdhu | Human Resources | East-233 |
1103 | NULL | randerss | Marketing | East-460 |
1109 | f229g533h679 | nlocklea | Sales | East-196 |
1113 | j781k420l510 | pjaimes | Human Resources | East-366 |
1121 | z628s557t397 | mrojas | Sales | East-288 |
1135 | z934g229h883 | khamamot | Information Technology | East-186 |
1140 | k982l199m839 | apatel2 | Human Resources | East-385 |
1142 | m674n127o823 | lsilva | Finance | East-440 |
1143 | n704o364p471 | sstark | Information Technology | East-282 |
1155 | z942a966b589 | zjones | Human Resources | East-122 |
1156 | a184b775c707 | dellery | Marketing | East-417 |
1159 | d881e710f732 | jshen | Finance | East-193 |
1161 | f951g408h866 | jjenkins | Information Technology | East-433 |
1163 | h679i515j339 | cwilliam | Marketing | East-216 |
1177 | w691u183x228 | aezra | Human Resources | East-190 |
1183 | b566c710d544 | lguraish | Human Resources | East-400 |
1187 | f963g637h851 | bhode | Finance | East-351 |
1192 | k570l183m949 | rlaghari | Information Technology | East-138 |
1195 | n516o853p957 | orainier | Finance | East-346 |
1199 | r520s571t459 | areyes | Human Resources | East-100 |

41 rows in set (0.001 sec)

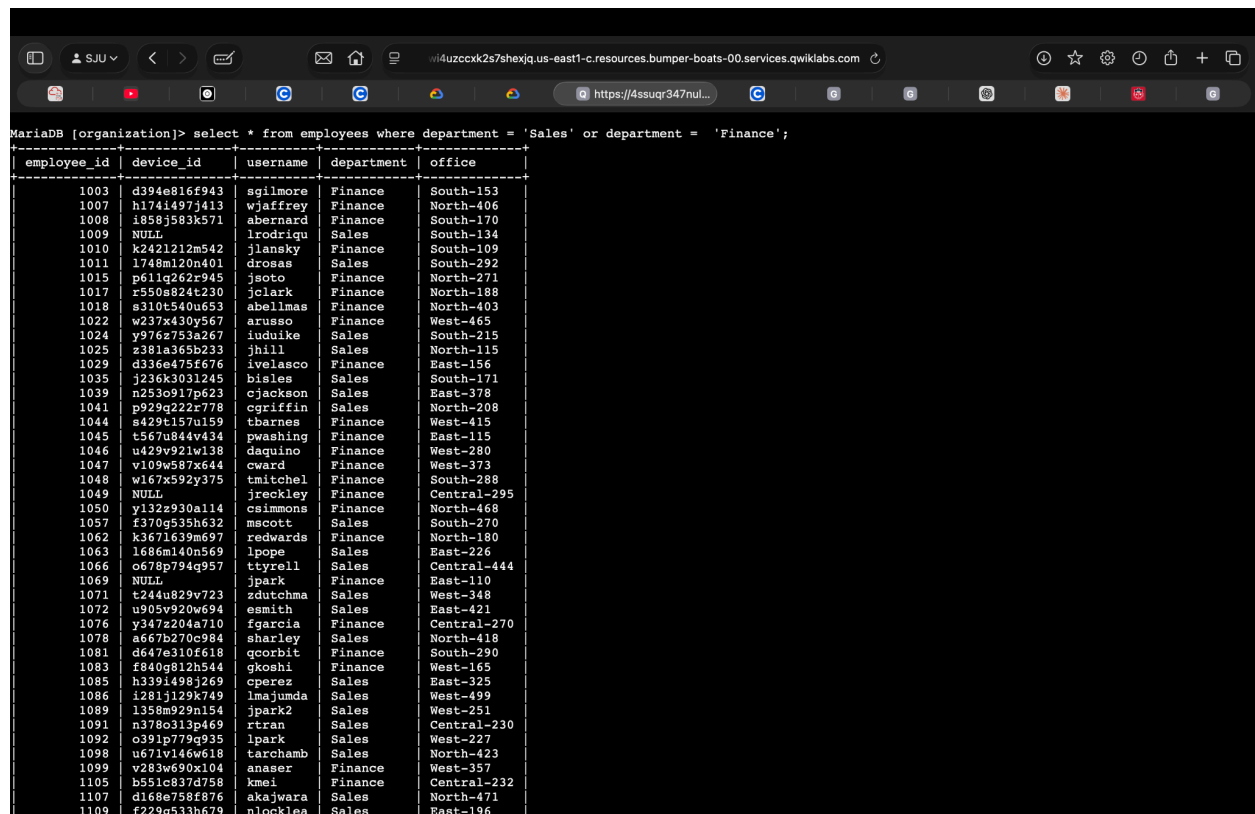
MariaDB [organization]> select *
-> from employees
-> where department = 'Marketing' and office like 'EAST%';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |
| 1103 | NULL | randerss | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
+-----+-----+-----+-----+-----+

7 rows in set (0.033 sec)

MariaDB [organization]>
```

## Retrieve employees in Finance or Sales

The security team required updates on the employee departments in the Sales and Finance departments. I inputted **SELECT \*** to look through all information categories and **FROM employees** to retrieve employee information. Then, I added the argument **WHERE department = 'Sales'** to query all departments in Sales. The **OR** argument includes the 2nd part of the argument that queries the Finance department. I made sure to include **department = 'Finance'** as that includes both Sales and Finance departments.



The screenshot shows a terminal window with a SQL query and its results. The query is: `select * from employees where department = 'Sales' or department = 'Finance';`. The results are displayed in a table with 5 columns: employee\_id, device\_id, username, department, and office. The table contains 40 rows of data, listing employees from both the Sales and Finance departments.

| employee_id | device_id    | username  | department | office      |
|-------------|--------------|-----------|------------|-------------|
| 1003        | d394e816f943 | sgillmore | Finance    | South-153   |
| 1007        | h1741497j413 | wjaffrey  | Finance    | North-406   |
| 1008        | i858j583k571 | abernard  | Finance    | South-170   |
| 1009        | NULL         | lrodriagu | Sales      | South-134   |
| 1010        | k2421212m542 | jlansky   | Finance    | South-109   |
| 1011        | 1748m120n401 | drosas    | Sales      | South-292   |
| 1015        | p611q262r945 | jsoto     | Finance    | North-271   |
| 1017        | r550s824t230 | jclark    | Finance    | North-188   |
| 1018        | s310t540u653 | abellmas  | Finance    | North-403   |
| 1022        | w237x430y567 | arusso    | Finance    | West-465    |
| 1024        | y976z753a267 | iuduike   | Sales      | South-215   |
| 1025        | z381a365b233 | jhill     | Sales      | North-115   |
| 1029        | d336e475f676 | ivelasgo  | Finance    | East-156    |
| 1035        | j236k303l245 | hisles    | Sales      | South-171   |
| 1039        | n253o917p623 | cjackson  | Sales      | East-378    |
| 1041        | p929q222r778 | cgriffin  | Sales      | North-208   |
| 1044        | s429t157u159 | tbarnes   | Finance    | West-415    |
| 1045        | t567u844v434 | pwashing  | Finance    | East-115    |
| 1046        | u429v921w138 | daquino   | Finance    | West-280    |
| 1047        | v109w587x644 | cward     | Finance    | West-373    |
| 1048        | w167x592y375 | tmitchel  | Finance    | South-288   |
| 1049        | NULL         | jreckley  | Finance    | Central-295 |
| 1050        | y132z930a114 | csimmons  | Finance    | North-468   |
| 1057        | f370g535h632 | mseott    | Sales      | South-270   |
| 1062        | k367l639m697 | redwards  | Finance    | North-180   |
| 1063        | l686m140n569 | lpope     | Sales      | East-226    |
| 1066        | o678p794q957 | ttyrell   | Sales      | Central-444 |
| 1069        | NULL         | jpark     | Finance    | East-110    |
| 1071        | t244u829v723 | zdutchma  | Sales      | West-348    |
| 1072        | u905v920w694 | esmith    | Sales      | East-421    |
| 1076        | y347z204a710 | fgarcia   | Finance    | Central-270 |
| 1078        | a667b270c984 | sharley   | Sales      | North-418   |
| 1081        | d647e310f618 | qcorbit   | Finance    | South-290   |
| 1083        | f840g812h544 | gvoshi    | Finance    | West-165    |
| 1085        | h339i498j269 | cperez    | Sales      | East-325    |
| 1086        | i281j129k749 | lmajumda  | Sales      | West-499    |
| 1089        | l358m929n154 | jpark2    | Sales      | West-251    |
| 1091        | n378o313p469 | rtran     | Sales      | Central-230 |
| 1092        | o391p779q935 | lpark     | Sales      | West-227    |
| 1098        | u671v146w618 | tarchamb  | Sales      | North-423   |
| 1099        | v283w690x104 | anaser    | Finance    | West-357    |
| 1105        | b551c837d758 | kmei      | Finance    | Central-232 |
| 1107        | d168e758f876 | akajwara  | Sales      | North-471   |
| 1109        | f229g533h679 | nlocklea  | Sales      | East-196    |

## Retrieve all employees not in IT

The security team required one more update for all departments excluding IT. IT had already received the update and all other departments required it. I created an SQL query beginning with **SELECT \*** and **FROM employees** again. Then, I added 2 arguments **WHERE NOT** to lookup departments that are not Information Technology. The **NOT** negates the accepted argument which will return every other department.

```
33 rows in set, 5 warnings (0.001 sec)
```

MariaDB [organization]> select \*  
-> from employees  
-> where not department = 'Information Technology';

| employee_id | device_id    | username  | department      | office      |
|-------------|--------------|-----------|-----------------|-------------|
| 1000        | a320b137c219 | el Larson | Marketing       | East-170    |
| 1001        | b239c825d303 | bmoreno   | Marketing       | Central-276 |
| 1002        | c116d593e558 | tshah     | Human Resources | North-434   |
| 1003        | d394e816f943 | sgilmore  | Finance         | South-153   |
| 1004        | e218f877g788 | eraab     | Human Resources | South-127   |
| 1005        | f551g340h864 | gesparza  | Human Resources | South-366   |
| 1007        | h174i497j413 | wjaffrey  | Finance         | North-406   |
| 1008        | i858j583k571 | abernard  | Finance         | South-170   |
| 1009        | NULL         | lrodrigu  | Sales           | South-134   |
| 1010        | k242l121m542 | jlansky   | Finance         | South-109   |
| 1011        | l748m120n401 | drozas    | Sales           | South-292   |
| 1015        | p611q262r945 | jsoto     | Finance         | North-271   |
| 1016        | q793r736s288 | sbaelish  | Human Resources | North-229   |
| 1017        | r550s824t230 | jclark    | Finance         | North-188   |
| 1018        | s310t540u653 | abellmas  | Finance         | North-403   |
| 1020        | u899v381w363 | arutley   | Marketing       | South-351   |
| 1022        | w237x430y567 | arusso    | Finance         | West-465    |
| 1024        | y976z753a267 | ludulke   | Sales           | South-215   |
| 1025        | z301a365b233 | jhill     | Sales           | North-115   |
| 1026        | a998b568c863 | spatel    | Human Resources | West-320    |
| 1027        | b806c503d354 | mrar      | Marketing       | West-246    |
| 1028        | c603d749e374 | astrada   | Human Resources | West-121    |
| 1029        | d336e475f676 | ivelasco  | Finance         | East-156    |
| 1030        | e391f189g913 | mabadi    | Marketing       | West-375    |
| 1031        | f419g188h578 | dkot      | Marketing       | West-408    |
| 1034        | i679j565k940 | bsand     | Human Resources | East-484    |
| 1035        | j236k303l245 | bisles    | Sales           | South-171   |
| 1036        | k550l53m205  | rjensen   | Marketing       | Central-239 |
| 1038        | m873n636o225 | btang     | Human Resources | Central-260 |
| 1039        | n253o917p623 | cjackson  | Sales           | East-378    |
| 1040        | o783p832q294 | dtarly    | Human Resources | East-237    |
| 1041        | p929q822r778 | cgriffin  | Sales           | North-208   |
| 1042        | q175r338s833 | acook     | Human Resources | West-381    |
| 1044        | s429t157u159 | tbarnes   | Finance         | West-415    |
| 1045        | t567u844v434 | pwashing  | Finance         | East-115    |
| 1046        | u429v921w138 | daquino   | Finance         | West-280    |
| 1047        | v109w587x644 | cward     | Finance         | West-373    |
| 1048        | w167x592y375 | tmitchel  | Finance         | South-288   |
| 1049        | NULL         | jreckley  | Finance         | Central-295 |
| 1050        | y132z930a114 | csimmons  | Finance         | North-468   |
| 1051        | z451a308b518 | itraora   | Marketing       | Central-134 |
| 1052        | a192b174c940 | jdarosa   | Marketing       | East-195    |
| 1053        | b979c871d361 | nemmanue  | Human Resources | Central-259 |

# Match employees to their machines

The department required machines to be matched to employees. This would work with an inner join which correctly returns columns that are needed based on columns. I queried this by using **SELECT \*** and **FROM machines**. Then, I used the new syntax of **INNER JOIN** and matched it with the employees table. Then, I added **ON machines.device\_id = employees.device\_id** to specify table 1 to join table 2 which is the employees table. This returned all employee information and machines that have matches to create a bigger table. The **device\_id** is a shared key which is the only information shared on both piles of data.

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
MariaDB [organization]> clear
MariaDB [organization]> clear
MariaDB [organization]> select *
-> from machines
-> inner join employees on machines.device_id = employees.device_id;
```

| device_id    | operating_system | email_client   | OS_patch_date | employee_id | employee_id | device_id    | username | department             | office      |
|--------------|------------------|----------------|---------------|-------------|-------------|--------------|----------|------------------------|-------------|
| a320b137c219 | OS 2             | Email Client 2 | 2021-03-01    | 1000        | 1000        | a320b137c219 | elarsen  | Marketing              | East-170    |
| b239c825d303 | OS 1             | Email Client 1 | 2021-03-01    | 1001        | 1001        | b239c825d303 | bmoreno  | Marketing              | Central-276 |
| c116d593e558 | OS 3             | Email Client 1 | 2021-09-01    | 1002        | 1002        | c116d593e558 | tshah    | Human Resources        | North-434   |
| d394e816f943 | OS 3             | Email Client 2 | 2021-03-01    | 1003        | 1003        | d394e816f943 | sgilmore | Finance                | South-153   |
| e218f877q788 | OS 2             | Email Client 1 | 2021-09-01    | 1004        | 1004        | e218f877q788 | eraab    | Human Resources        | South-127   |
| f551g340h864 | OS 3             | Email Client 2 | 2021-12-01    | 1005        | 1005        | f551g340h864 | gesparza | Human Resources        | South-366   |
| g329h357i597 | OS 1             | Email Client 2 | 2021-06-01    | 1006        | 1006        | g329h357i597 | alevitsk | Information Technology | East-320    |
| h174i497j413 | OS 2             | Email Client 1 | 2021-03-01    | 1007        | 1007        | h174i497j413 | wjaffrey | Finance                | North-406   |
| i858j583k571 | OS 2             | Email Client 2 | 2021-06-01    | 1008        | 1008        | i858j583k571 | abernard | Finance                | South-170   |
| k242l212m542 | OS 1             | Email Client 1 | 2021-03-01    | 1010        | 1010        | k242l212m542 | jlansky  | Finance                | South-109   |
| l748m120n401 | OS 3             | Email Client 1 | 2021-09-01    | 1011        | 1011        | l748m120n401 | droasa   | Sales                  | South-292   |
| m756n668o146 | OS 1             | Email Client 2 | 2021-09-01    | 1012        | 1012        | m756n668o146 | nmason   | Information Technology | North-160   |
| n205o559p243 | OS 1             | Email Client 2 | 2021-06-01    | 1013        | 1013        | n205o559p243 | zbernal  | Information Technology | South-229   |
| p611q262r945 | OS 1             | Email Client 2 | 2021-03-01    | 1015        | 1015        | p611q262r945 | jsoto    | Finance                | North-271   |
| q793r736s288 | OS 2             | Email Client 1 | 2021-03-01    | 1016        | 1016        | q793r736s288 | sbaelish | Human Resources        | North-229   |
| r550s824t230 | OS 2             | Email Client 1 | 2021-03-01    | 1017        | 1017        | r550s824t230 | jclark   | Finance                | North-188   |
| s310t540u653 | OS 1             | Email Client 2 | 2021-12-01    | 1018        | 1018        | s310t540u653 | abellmas | Finance                | North-403   |
| t815u205v470 | OS 3             | Email Client 1 | 2021-06-01    | 1019        | 1019        | t815u205v470 | mcouliba | Information Technology | North-108   |
| u899v381w363 | OS 3             | Email Client 1 | 2021-06-01    | 1020        | 1020        | u899v381w363 | arutley  | Marketing              | South-351   |
| v200w121x977 | OS 2             | Email Client 2 | 2021-03-01    | 1021        | 1021        | v200w121x977 | smartell | Information Technology | South-138   |
| w237x430y567 | OS 2             | Email Client 2 | 2021-06-01    | 1022        | 1022        | w237x430y567 | aruaeo   | Finance                | West-465    |
| x253y759z103 | OS 1             | Email Client 2 | 2021-12-01    | 1023        | 1023        | x253y759z103 | aalonso  | Information Technology | West-393    |
| y976z753a267 | OS 2             | Email Client 2 | 2021-06-01    | 1024        | 1024        | y976z753a267 | iuduke   | Sales                  | South-215   |
| z381a365b233 | OS 3             | Email Client 2 | 2021-12-01    | 1025        | 1025        | z381a365b233 | jhill    | Sales                  | North-115   |
| a998b568c863 | OS 3             | Email Client 1 | 2021-12-01    | 1026        | 1026        | a998b568c863 | apatel   | Human Resources        | West-320    |
| b806c503d354 | OS 2             | Email Client 1 | 2021-12-01    | 1027        | 1027        | b806c503d354 | mrah     | Marketing              | West-246    |
| c603d749e374 | OS 1             | Email Client 1 | 2021-12-01    | 1028        | 1028        | c603d749e374 | astrada  | Human Resources        | West-121    |
| d336e475f676 | OS 2             | Email Client 2 | 2021-06-01    | 1029        | 1029        | d336e475f676 | ivelasco | Finance                | East-156    |
| e391f189g913 | OS 3             | Email Client 2 | 2021-12-01    | 1030        | 1030        | e391f189g913 | mabadi   | Marketing              | West-375    |
| f419g188h578 | OS 1             | Email Client 1 | 2021-06-01    | 1031        | 1031        | f419g188h578 | dkot     | Marketing              | West-408    |
| g773h303i639 | OS 2             | Email Client 2 | 2021-12-01    | 1032        | 1032        | g773h303i639 | jrafasl  | Information Technology | Central-309 |
| i679j565k940 | OS 1             | Email Client 1 | 2021-03-01    | 1034        | 1034        | i679j565k940 | bsand    | Human Resources        | East-484    |
| j236k303l245 | OS 1             | Email Client 1 | 2021-12-01    | 1035        | 1035        | j236k303l245 | bisles   | Sales                  | South-171   |
| k550l533m205 | OS 3             | Email Client 2 | 2021-09-01    | 1036        | 1036        | k550l533m205 | rjensen  | Marketing              | Central-239 |
| l693m585n528 | OS 3             | Email Client 1 | 2021-06-01    | 1037        | 1037        | l693m585n528 | dtanaka  | Information Technology | West-468    |
| m873n636o225 | OS 1             | Email Client 2 | 2021-06-01    | 1038        | 1038        | m873n636o225 | btang    | Human Resources        | Central-260 |
| n253o917p623 | OS 1             | Email Client 2 | 2021-09-01    | 1039        | 1039        | n253o917p623 | cjackson | Sales                  | East-378    |
| o783p832q294 | OS 2             | Email Client 2 | 2021-09-01    | 1040        | 1040        | o783p832q294 | dtarly   | Human Resources        | East-237    |
| p929q222r778 | OS 3             | Email Client 1 | 2021-06-01    | 1041        | 1041        | p929q222r778 | cgriffin | Sales                  | North-208   |
| q175r338s833 | OS 3             | Email Client 1 | 2021-06-01    | 1042        | 1042        | q175r338s833 | acook    | Human Resources        | West-381    |
| s429t157u159 | OS 1             | Email Client 1 | 2021-03-01    | 1044        | 1044        | s429t157u159 | tbarnes  | Finance                | West-415    |

## Return information on employees and machines

The organization asks me to find every device that is unassigned in the closet. If a machine has no user then it will return as null. A left join is where every row from the first table is kept even if there is no second table. Again, I begin the query with **SELECT \*** and **FROM machines**. I use a **LEFT JOIN** on employees using **device\_id** from both machines and employees table. This returns every single employee regardless if they were assigned a device.

```
MariaDB [organization]> select *
-> from machines
-> left join employees on machines.device_id = employees.device_id;
```

| device_id    | operating_system | email_client   | OS_patch_date | employee_id | employee_id | device_id    | username | department             | office      |
|--------------|------------------|----------------|---------------|-------------|-------------|--------------|----------|------------------------|-------------|
| a320b137c219 | OS 2             | Email Client 2 | 2021-03-01    | 1000        | 1000        | a320b137c219 | elarson  | Marketing              | East-170    |
| b239c825d303 | OS 1             | Email Client 1 | 2021-03-01    | 1001        | 1001        | b239c825d303 | bmoreno  | Marketing              | Central-276 |
| c116d593e558 | OS 3             | Email Client 1 | 2021-09-01    | 1002        | 1002        | c116d593e558 | tshah    | Human Resources        | North-434   |
| d394e816f943 | OS 3             | Email Client 2 | 2021-03-01    | 1003        | 1003        | d394e816f943 | sgilmore | Finance                | South-153   |
| e218f877g788 | OS 2             | Email Client 1 | 2021-09-01    | 1004        | 1004        | e218f877g788 | eraab    | Human Resources        | South-127   |
| f551g340h864 | OS 3             | Email Client 2 | 2021-12-01    | 1005        | 1005        | f551g340h864 | gesparza | Human Resources        | South-366   |
| g329h357i597 | OS 1             | Email Client 2 | 2021-06-01    | 1006        | 1006        | g329h357i597 | alevitak | Information Technology | East-320    |
| h174i497j413 | OS 2             | Email Client 1 | 2021-03-01    | 1007        | 1007        | h174i497j413 | wjaffrey | Finance                | North-406   |
| i858j583k571 | OS 2             | Email Client 2 | 2021-06-01    | 1008        | 1008        | i858j583k571 | abernard | Finance                | South-170   |
| k242l121m542 | OS 1             | Email Client 1 | 2021-03-01    | 1010        | 1010        | k242l121m542 | jlansky  | Finance                | South-109   |
| l748m120n401 | OS 3             | Email Client 1 | 2021-09-01    | 1011        | 1011        | l748m120n401 | drosas   | Sales                  | South-292   |
| m756n668o146 | OS 1             | Email Client 2 | 2021-09-01    | 1012        | 1012        | m756n668o146 | nmason   | Information Technology | North-160   |
| n205o559p243 | OS 1             | Email Client 2 | 2021-06-01    | 1013        | 1013        | n205o559p243 | zbernal  | Information Technology | South-229   |
| p611q262r945 | OS 1             | Email Client 2 | 2021-03-01    | 1015        | 1015        | p611q262r945 | jsofo    | Finance                | North-271   |
| q793r736s288 | OS 2             | Email Client 1 | 2021-03-01    | 1016        | 1016        | q793r736s288 | sbaelish | Human Resources        | North-229   |
| r550e824t230 | OS 2             | Email Client 1 | 2021-03-01    | 1017        | 1017        | r550e824t230 | jclark   | Finance                | North-188   |
| s310t540u653 | OS 1             | Email Client 2 | 2021-12-01    | 1018        | 1018        | s310t540u653 | abellmas | Finance                | North-403   |
| t815u205v470 | OS 3             | Email Client 1 | 2021-06-01    | 1019        | 1019        | t815u205v470 | mcouliba | Information Technology | North-108   |
| u899v381w363 | OS 3             | Email Client 1 | 2021-06-01    | 1020        | 1020        | u899v381w363 | arutley  | Marketing              | South-351   |
| v200w121x977 | OS 2             | Email Client 2 | 2021-03-01    | 1021        | 1021        | v200w121x977 | smartell | Information Technology | South-138   |
| w237x430y567 | OS 2             | Email Client 2 | 2021-06-01    | 1022        | 1022        | w237x430y567 | arusso   | Finance                | West-465    |
| x253y759z103 | OS 1             | Email Client 2 | 2021-12-01    | 1023        | 1023        | x253y759z103 | aalonso  | Information Technology | West-393    |
| y976z753a267 | OS 2             | Email Client 2 | 2021-06-01    | 1024        | 1024        | y976z753a267 | iudulke  | Sales                  | South-215   |
| z381a365b233 | OS 3             | Email Client 2 | 2021-12-01    | 1025        | 1025        | z381a365b233 | jhill    | Sales                  | North-115   |
| a998b568c863 | OS 3             | Email Client 1 | 2021-12-01    | 1026        | 1026        | a998b568c863 | spatel   | Human Resources        | West-320    |
| b806c503d354 | OS 2             | Email Client 1 | 2021-12-01    | 1027        | 1027        | b806c503d354 | mrah     | Marketing              | West-246    |
| c603d749e374 | OS 1             | Email Client 1 | 2021-12-01    | 1028        | 1028        | c603d749e374 | aestrada | Human Resources        | West-121    |
| d336e475f676 | OS 2             | Email Client 2 | 2021-06-01    | 1029        | 1029        | d336e475f676 | ivelasco | Finance                | East-156    |
| e391f189g913 | OS 3             | Email Client 2 | 2021-12-01    | 1030        | 1030        | e391f189g913 | mabadi   | Marketing              | West-375    |
| f419g188h578 | OS 1             | Email Client 1 | 2021-06-01    | 1031        | 1031        | f419g188h578 | dkot     | Marketing              | West-408    |
| g773h303i639 | OS 2             | Email Client 2 | 2021-12-01    | 1032        | 1032        | g773h303i639 | jrafael  | Information Technology | Central-309 |
| i679j565k940 | OS 1             | Email Client 1 | 2021-03-01    | 1034        | 1034        | i679j565k940 | bsand    | Human Resources        | East-484    |
| j236k303l245 | OS 1             | Email Client 1 | 2021-12-01    | 1035        | 1035        | j236k303l245 | bisles   | Sales                  | South-171   |
| k550l533m205 | OS 3             | Email Client 2 | 2021-09-01    | 1036        | 1036        | k550l533m205 | rjensen  | Marketing              | Central-239 |
| l693m585n528 | OS 3             | Email Client 1 | 2021-06-01    | 1037        | 1037        | l693m585n528 | dtanaka  | Information Technology | West-468    |
| m873n636o225 | OS 1             | Email Client 2 | 2021-06-01    | 1038        | 1038        | m873n636o225 | btang    | Human Resources        | Central-260 |
| n253o917p623 | OS 1             | Email Client 2 | 2021-09-01    | 1039        | 1039        | n253o917p623 | cjackson | Sales                  | East-378    |
| o783p832q294 | OS 2             | Email Client 2 | 2021-09-01    | 1040        | 1040        | o783p832q294 | dtarly   | Human Resources        | East-237    |
| p929q222r778 | OS 3             | Email Client 1 | 2021-06-01    | 1041        | 1041        | p929q222r778 | cgriffin | Sales                  | North-208   |
| q175r338s833 | OS 3             | Email Client 1 | 2021-06-01    | 1042        | 1042        | q175r338s833 | acook    | Human Resources        | West-381    |
| s429t157u159 | OS 1             | Email Client 1 | 2021-03-01    | 1044        | 1044        | s429t157u159 | tbarnes  | Finance                | West-415    |
| t567u844v434 | OS 3             | Email Client 2 | 2021-12-01    | 1045        | 1045        | t567u844v434 | pwashing | Finance                | East-115    |
| u429v921w138 | OS 2             | Email Client 2 | 2021-12-01    | 1046        | 1046        | u429v921w138 | daquino  | Finance                | West-280    |
| v109w587x644 | OS 2             | Email Client 2 | 2021-03-01    | 1047        | 1047        | v109w587x644 | cward    | Finance                | West-373    |

The organization asks to see all employees on even if they haven't been issued a laptop. If an employee has no machine then a null value will return from the lookup. A right join keeps every single row on the second table even if there is no match on the first. In this query, I used a right join on employees. I began the query with **SELECT \*** and **FROM machines**. Then, I added the **RIGHT JOIN** employees syntax **ON machines.device\_id**. I made it **= to employees.device\_id** which matched the shared row. This returned all employees regardless of if they have a device or not.

MariaDB [organization]> select \*  
-> from machines  
-> right join employees on machines.device\_id = employees.device\_id;

| device_id    | operating_system | email_client   | OS_patch_date | employee_id | employee_id | device_id    | username  | department             | office      |
|--------------|------------------|----------------|---------------|-------------|-------------|--------------|-----------|------------------------|-------------|
| a320b137c219 | OS 2             | Email Client 2 | 2021-03-01    | 1000        | 1000        | a320b137c219 | elarson   | Marketing              | East-170    |
| b239c825d303 | OS 1             | Email Client 1 | 2021-03-01    | 1001        | 1001        | b239c825d303 | bmoreno   | Marketing              | Central-276 |
| c116d593e558 | OS 3             | Email Client 1 | 2021-09-01    | 1002        | 1002        | c116d593e558 | tshah     | Human Resources        | North-434   |
| d394e816f943 | OS 3             | Email Client 2 | 2021-03-01    | 1003        | 1003        | d394e816f943 | sgilmore  | Finance                | South-153   |
| e218f877g788 | OS 2             | Email Client 1 | 2021-09-01    | 1004        | 1004        | e218f877g788 | eraab     | Human Resources        | South-127   |
| f551g340h864 | OS 3             | Email Client 2 | 2021-12-01    | 1005        | 1005        | f551g340h864 | gesparza  | Human Resources        | South-366   |
| g329h357i597 | OS 1             | Email Client 2 | 2021-06-01    | 1006        | 1006        | g329h357i597 | alevitsk  | Information Technology | East-320    |
| h174i497j413 | OS 2             | Email Client 1 | 2021-03-01    | 1007        | 1007        | h174i497j413 | wjaffrey  | Finance                | North-406   |
| i858j583k571 | OS 2             | Email Client 2 | 2021-06-01    | 1008        | 1008        | i858j583k571 | abernard  | Finance                | South-170   |
| NULL         | NULL             | NULL           | NULL          | 1009        | 1009        | NULL         | lrodriagu | Sales                  | South-134   |
| k242l212m542 | OS 1             | Email Client 1 | 2021-03-01    | 1010        | 1010        | k242l212m542 | jlansky   | Finance                | South-109   |
| l748m120n401 | OS 3             | Email Client 1 | 2021-09-01    | 1011        | 1011        | l748m120n401 | drozas    | Sales                  | South-292   |
| m756n668o146 | OS 1             | Email Client 2 | 2021-09-01    | 1012        | 1012        | m756n668o146 | nmason    | Information Technology | North-160   |
| n205o559p243 | OS 1             | Email Client 2 | 2021-06-01    | 1013        | 1013        | n205o559p243 | zbernal   | Information Technology | South-229   |
| NULL         | NULL             | NULL           | NULL          | 1014        | 1014        | NULL         | asundara  | Information Technology | West-219    |
| p611q262r945 | OS 1             | Email Client 2 | 2021-03-01    | 1015        | 1015        | p611q262r945 | jsoto     | Finance                | North-271   |
| q793r736s288 | OS 2             | Email Client 1 | 2021-03-01    | 1016        | 1016        | q793r736s288 | sbaelish  | Human Resources        | North-229   |
| r550s824t230 | OS 2             | Email Client 1 | 2021-03-01    | 1017        | 1017        | r550s824t230 | jclark    | Finance                | North-188   |
| s310t540u653 | OS 1             | Email Client 2 | 2021-12-01    | 1018        | 1018        | s310t540u653 | abellmas  | Finance                | North-403   |
| t815u205v470 | OS 3             | Email Client 1 | 2021-06-01    | 1019        | 1019        | t815u205v470 | mcouliba  | Information Technology | North-108   |
| u899v381w363 | OS 3             | Email Client 1 | 2021-06-01    | 1020        | 1020        | u899v381w363 | arutley   | Marketing              | South-351   |
| v200w121x977 | OS 2             | Email Client 2 | 2021-03-01    | 1021        | 1021        | v200w121x977 | smartell  | Information Technology | South-138   |
| w237x430y567 | OS 2             | Email Client 2 | 2021-06-01    | 1022        | 1022        | w237x430y567 | arusso    | Finance                | West-465    |
| x253y759z103 | OS 1             | Email Client 2 | 2021-12-01    | 1023        | 1023        | x253y759z103 | aalonso   | Information Technology | West-393    |
| y976z753a267 | OS 2             | Email Client 2 | 2021-06-01    | 1024        | 1024        | y976z753a267 | iuduike   | Sales                  | South-215   |
| z381a365b233 | OS 3             | Email Client 2 | 2021-12-01    | 1025        | 1025        | z381a365b233 | jhill     | Sales                  | North-115   |
| a998b568c863 | OS 3             | Email Client 1 | 2021-12-01    | 1026        | 1026        | a998b568c863 | apatel    | Human Resources        | West-320    |
| b806c503d354 | OS 2             | Email Client 1 | 2021-12-01    | 1027        | 1027        | b806c503d354 | mrah      | Marketing              | West-246    |
| c603d749e374 | OS 1             | Email Client 1 | 2021-12-01    | 1028        | 1028        | c603d749e374 | aestrada  | Human Resources        | West-121    |
| d336e475f676 | OS 2             | Email Client 2 | 2021-06-01    | 1029        | 1029        | d336e475f676 | ivelasco  | Finance                | East-156    |
| e391f189g913 | OS 3             | Email Client 2 | 2021-12-01    | 1030        | 1030        | e391f189g913 | mabadii   | Marketing              | West-375    |
| f419g188h578 | OS 1             | Email Client 1 | 2021-06-01    | 1031        | 1031        | f419g188h578 | dkot      | Marketing              | West-408    |
| g773h303i639 | OS 2             | Email Client 2 | 2021-12-01    | 1032        | 1032        | g773h303i639 | jrafael   | Information Technology | Central-309 |
| NULL         | NULL             | NULL           | NULL          | 1033        | 1033        | NULL         | yappiah   | Information Technology | West-387    |
| i679j565k940 | OS 1             | Email Client 1 | 2021-03-01    | 1034        | 1034        | i679j565k940 | bsand     | Human Resources        | East-484    |
| j236k303l245 | OS 1             | Email Client 1 | 2021-12-01    | 1035        | 1035        | j236k303l245 | bisles    | Sales                  | South-171   |
| k550l533m205 | OS 3             | Email Client 2 | 2021-09-01    | 1036        | 1036        | k550l533m205 | rjensen   | Marketing              | Central-239 |
| l693m585n528 | OS 3             | Email Client 1 | 2021-06-01    | 1037        | 1037        | l693m585n528 | dtanaka   | Information Technology | West-468    |
| m873n636o225 | OS 1             | Email Client 2 | 2021-06-01    | 1038        | 1038        | m873n636o225 | btang     | Human Resources        | Central-260 |
| n253o917p623 | OS 1             | Email Client 2 | 2021-09-01    | 1039        | 1039        | n253o917p623 | cjackson  | Sales                  | East-378    |
| o782p932q294 | OS 2             | Email Client 2 | 2021-09-01    | 1040        | 1040        | o782p932q294 | dtarily   | Human Resources        | East-237    |
| p929q222r778 | OS 3             | Email Client 1 | 2021-06-01    | 1041        | 1041        | p929q222r778 | cgriffin  | Sales                  | North-208   |
| q175r338s833 | OS 3             | Email Client 1 | 2021-06-01    | 1042        | 1042        | q175r338s833 | acook     | Human Resources        | West-381    |
| NULL         | NULL             | NULL           | NULL          | 1043        | 1043        | NULL         | lyamamot  | Information Technology | East-354    |

## Retrieve login attempt data

To continue the investigation on the security incident, a query regarding information on all employees who made login attempts was required. An inner join was used for this process. I began the query with **SELECT \*** and **FROM employees**. Then, I added **INNER JOIN log\_in\_attempts** to garner data on login attempts. Then, I added on **employees.username = log\_in\_attempts.username** to link common usernames on the database. This allowed me to view all IP addresses, successful logins, and the username of the user that logged in at a specific time.

Personal

y4q5fz7a3glkcpza.us-west1-b.resources.bumper-boats-00.services.qwiklabs.com

MariaDB [organization]> clear

MariaDB [organization]> clear

MariaDB [organization]> select \*

-> from employees

-> inner join log\_in\_attempts on employees.username = log\_in\_attempts.username;

|   | employee_id | device_id    | username | department             | office      | event_id | username | login_date | login_time | country | ip_address      | success |
|---|-------------|--------------|----------|------------------------|-------------|----------|----------|------------|------------|---------|-----------------|---------|
| 0 | 1032        | g773h303i639 | jrafael  | Information Technology | Central-309 | 1        | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 |         |
| 0 | 1026        | a998b568c863 | apatel   | Human Resources        | West-320    | 2        | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |         |
| 0 | 1031        | f419g188h578 | dkot     | Marketing              | West-408    | 3        | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 |         |
| 0 | 1031        | f419g188h578 | dkot     | Marketing              | West-408    | 4        | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  |         |
| 0 | 1032        | g773h303i639 | jrafael  | Information Technology | Central-309 | 5        | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  |         |
| 0 | 1020        | u899v381w363 | arutley  | Marketing              | South-351   | 6        | arutley  | 2022-05-12 | 17:00:59   | MEXICO  | 192.168.3.24    |         |
| 0 | 1004        | e218f877g788 | eraab    | Human Resources        | South-127   | 7        | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 |         |
| 0 | 1035        | j236k303l245 | bisles   | Sales                  | South-171   | 8        | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 |         |
| 0 | 1033        | NULL         | yappiah  | Information Technology | West-387    | 9        | yappiah  | 2022-05-11 | 13:47:29   | MEX     | 192.168.59.136  |         |
| 0 | 1032        | g773h303i639 | jrafael  | Information Technology | Central-309 | 10       | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 |         |
| 0 | 1003        | d394e816f943 | sgilmore | Finance                | South-153   | 11       | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  |         |
| 0 | 1031        | f419g188h578 | dkot     | Marketing              | West-408    | 12       | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 |         |
| 0 | 1027        | b806c503d354 | mrah     | Marketing              | West-246    | 13       | mrah     | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 |         |
| 0 | 1016        | q793r736s288 | sbaelish | Human Resources        | North-229   | 14       | sbaelish | 2022-05-10 | 10:20:18   | US      | 192.168.16.99   |         |
| 0 | 1043        | NULL         | lyamamot | Information Technology | East-354    | 15       | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  |         |
| 0 | 1019        | t815u205v470 | mcouliba | Information Technology | North-108   | 16       | mcouliba | 2022-05-11 | 06:44:22   | CAN     | 192.168.172.189 |         |
| 0 | 1045        | t567u844v434 | pwashing | Finance                | East-115    | 17       | pwashing | 2022-05-11 | 02:33:02   | USA     | 192.168.81.89   |         |
| 0 | 1045        | t567u844v434 | pwashing | Finance                | East-115    | 18       | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |         |
| 0 | 1025        | z381a365b233 | jhill    | Sales                  | North-115   | 19       | jhill    | 2022-05-12 | 13:09:04   | US      | 192.168.142.245 |         |
| 0 | 1002        | c116d593e558 | tshah    | Human Resources        | North-434   | 20       | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  |         |

## Summary

Overall, this was a great introduction experience into SQL and database queries. Using different arguments to filter each query gave insight into how real-world database environments are utilized. The different arguments have purposes and garner information uniquely. This showed the significance of data efficiency as without adding filters would make data lookup time consuming.